# Predictive Analytics for Smart City Infrastructure

M.Tech Project Work Final Review

Presented by

**S K Sricharan (25MML1010)**

M.Tech Computer Science and Engineering

(Artificial Intelligence and Machine Learning – In collaboration with LTI Mindtree)

Under the Guidance of

Dr R. Srinivasa Perumal

Associate Professor

School of Computer Science and Engineering

Vellore Institute of Technology, Chennai.

# CONTENTS


- ABSTRACT
- INTRODUCTION
- LITERATURE SURVEY
- RESEARCH GAPS
- OBJECTIVES
- RESEARCH WORK DONE
  - MODULE IDENTIFICATION
  - SYSTEM ARCHITECTURE
  - DEVELOPMENT & INTEGRATION
  - PROTOTYPE / RESEARCH FRAMEWORK
- CONCLUSIONS
- OUTCOME
- RESULTS & OUTCOMES
- REFERENCES

# 1.ABSTRACT

Traffic control rooms can see congestion as it happens, but rarely receive forecasts that are explained, honest about their uncertainty and tested before an action is suggested. This project builds a decision-support prototype on public New York City data. A context-gated GATv2–Transformer forecasts link speeds 15, 30 and 60 minutes ahead on a graph of 16 Manhattan DOT speed links (January–October 2026), fusing weather, 311 and calendar context through an availability-aware gate. Split-conformal intervals quantify uncertainty, grouped SHAP explains the trained model, and a LangGraph workflow uses a local LLM (qwen2.5 7B) with FAISS retrieval to draft signal advisories that deterministic safety rules and a SUMO simulation must pass before human review. On a 45-day test period the model reaches 4.77, 5.33 and 6.15 mph MAE, 7–16 % below persistence; intervals reach 90 % coverage overall, and retrieval removed every fabricated citation. An audit of the simulation testbed showed it cannot yet support a controller comparison, which is therefore reported as not run.

# 2. INTRODUCTION

- Urban congestion depends on demand, neighbouring roads, weather, events and incidents, yet control rooms mostly see only current conditions.
- Graph neural networks forecast traffic on road networks [1]–[3], but give point forecasts without uncertainty or explanation.
- An LLM can phrase advice convincingly without evidence; retrieval grounding [4] and simulation testing [5] are needed before any signal action.
- Objective: join forecasts, uncertainty, explanations, cited evidence and simulation checks in one auditable, human-in-the-loop advisory pipeline on real NYC data.

[1] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in Proc. ICLR, 2018.
[2] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in Proc. IJCAI, 2018, pp. 3634–3640.
[3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, “Graph WaveNet for deep spatial-temporal graph modeling,” in Proc. IJCAI, 2019, pp. 1907–1913.
[4] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Proc. NeurIPS, 2020.
[5] P. A. Lopez et al., “Microscopic traffic simulation using SUMO,” in Proc. IEEE ITSC, 2018, pp. 2575–2582.

# 3.LITERATURE SURVEY

| S.No | Article Title | Techniques Used | Key Findings | Limitations |
|---|---|---|---|---|
| 1 | [1] Li et al., Diffusion Convolutional Recurrent Neural Network (ICLR 2018) | Diffusion convolution on a directed road graph inside a GRU encoder–decoder (seq2seq) | Modelling traffic as diffusion on the directed graph improved multi-step speed forecasts over non-graph baselines | Fixed, predefined graph; point forecasts only; no exogenous context or uncertainty |
| 2 | [2] Yu et al., Spatio-Temporal Graph Convolutional Networks (IJCAI 2018) | Graph convolution combined with gated temporal convolution; fully convolutional, no recurrence | Accurate short-term forecasts with faster training and fewer parameters than recurrent models | Static graph; point forecasts; no weather or event context |
| 3 | [3] Wu et al., Graph WaveNet (IJCAI 2019) | Self-adaptive adjacency matrix learned from data plus dilated causal convolutions | Captures hidden spatial dependencies even when the given graph is incomplete; long receptive field | Learned graph is hard to interpret; no uncertainty estimates |
| 4 | [6] Brody et al., How Attentive are Graph Attention Networks? (ICLR 2022) | GATv2: reorders the attention computation so that attention becomes dynamic | Shows that GAT attention is static; GATv2 is strictly more expressive and performs better on benchmarks | General-purpose graph layer; no temporal modelling |


[1] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in Proc. ICLR, 2018.
[2] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in Proc. IJCAI, 2018, pp. 3634–3640.
[3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, “Graph WaveNet for deep spatial-temporal graph modeling,” in Proc. IJCAI, 2019, pp. 1907–1913.
[6] S. Brody, U. Alon, and E. Yahav, “How attentive are graph attention networks?,” in Proc. ICLR, 2022.

# 3.LITERATURE SURVEY (CONTD.)

| S.No | Article Title | Techniques Used | Key Findings | Limitations |
|---|---|---|---|---|
| 5 | [7] Angelopoulos & Bates, A Gentle Introduction to Conformal Prediction (2021) | Split conformal prediction wrapped around any trained model | Distribution-free, finite-sample marginal coverage guarantee | Assumes exchangeable data; guarantees marginal, not conditional, coverage |
| 6 | [8] Xu & Xie, Conformal Prediction Interval for Dynamic Time-Series (ICML 2021) | EnbPI: bootstrap ensemble with conformal-style intervals for time series | Approximately valid intervals without exchangeability, under stated assumptions on the errors | Requires an ensemble of models; validity is approximate |
| 7 | [9] Lundberg & Lee, A Unified Approach to Interpreting Model Predictions (NeurIPS 2017) | SHAP: Shapley-value feature attribution (KernelSHAP and model-specific variants) | Unifies additive attribution methods with consistency guarantees | Cost grows with the number of features; attributions describe the model, not causes |
| 8 | [4] Lewis et al., Retrieval-Augmented Generation (NeurIPS 2020) | Dense passage retriever combined with a seq2seq generator | More specific and factual generation on knowledge-intensive tasks than a parametric-only model | Bounded by corpus and retriever quality; can still produce unsupported text |

[4] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Proc. NeurIPS, 2020.
[7] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” arXiv:2107.07511, 2021.
[8] C. Xu and Y. Xie, “Conformal prediction interval for dynamic time-series,” in Proc. ICML, PMLR vol. 139, 2021, pp. 11559–11569.
[9] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Proc. NeurIPS, 2017.

# 4. RESEARCH GAPS

### G1 Forecast stops at the number

Graph forecasting studies [1]–[3] report accuracy but do not connect the forecast to a checked, testable action.

### G2 Uncertainty does not decide

Point forecasts dominate; interval width and data quality rarely decide whether an advisory is issued.

### G3 Surrogate explanations

Explanations are often computed on simpler surrogate models instead of the deployed forecaster.

### G4 Ungrounded LLM advice

LLM recommendations are rarely tied to cited, allow-listed evidence or bounded by deterministic safety rules.

### G5 Missing context is hidden

Dead sensors and unavailable feeds are usually imputed silently instead of being modelled with availability masks.

### G6 Untested testbeds

Simulation studies seldom audit route choice, demand and signal timing before comparing controllers.

# 5. OBJECTIVES

**O1** Forecast link speeds 15, 30 and 60 min ahead with a context-gated GATv2–Transformer and compare it with six baselines on identical data

Addresses: G1, G5

**O2** Quantify uncertainty with per-horizon split-conformal intervals and defer advisories when reliability is low

Addresses: G2

**O3** Explain the trained model itself with grouped SHAP and sensitivity tests

Addresses: G3

**O4** Draft advisories with a local LLM that must cite retrieved evidence and pass deterministic safety checks

Addresses: G4

**O5** Test admissible advisories in a SUMO simulation, after verifying the testbed itself

Addresses: G6

**O6** Make every result reproducible and traceable: tests, MLflow, Docker, evidence registry and a paper generated from artifacts

Addresses: all

# 6. RESEARCH WORK DONE

- The research work is organised in four parts:
  - ❖ **MODULE IDENTIFICATION**
  - ❖ **SYSTEM ARCHITECTURE**
  - ❖ **DEVELOPMENT & INTEGRATION**
  - ❖ **PROTOTYPE / RESEARCH FRAMEWORK**

**1,968,303**
raw speed records processed

**16**
Manhattan links modelled

**7**
forecasting models compared

**95**
automated tests passing

# MODULE IDENTIFICATION

### M1 Data ingestion
NYC Open Data, Open-Meteo, OSM; resumable downloads with SHA-256 provenance

### M2 Preprocessing & features
5-minute grid, causal lags, availability masks, leakage-tested splits

### M3 Road graph
Directed DOT-link graph (26 nodes, 16 edges) as a PyG Temporal signal [10]

### M4 Forecasting
Context-gated GATv2–Transformer plus six baselines at 15/30/60 min

### M5 Uncertainty
Per-horizon split-conformal intervals and a reliability gate

### M6 Explainability
Grouped KernelSHAP on the trained model; neighbour and window sensitivity

### M7 Knowledge & RAG
15 approved documents, MiniLM embeddings, FAISS index

### M8 LLM agent
LangGraph workflow; local qwen2.5 7B via Ollama; schema-validated JSON

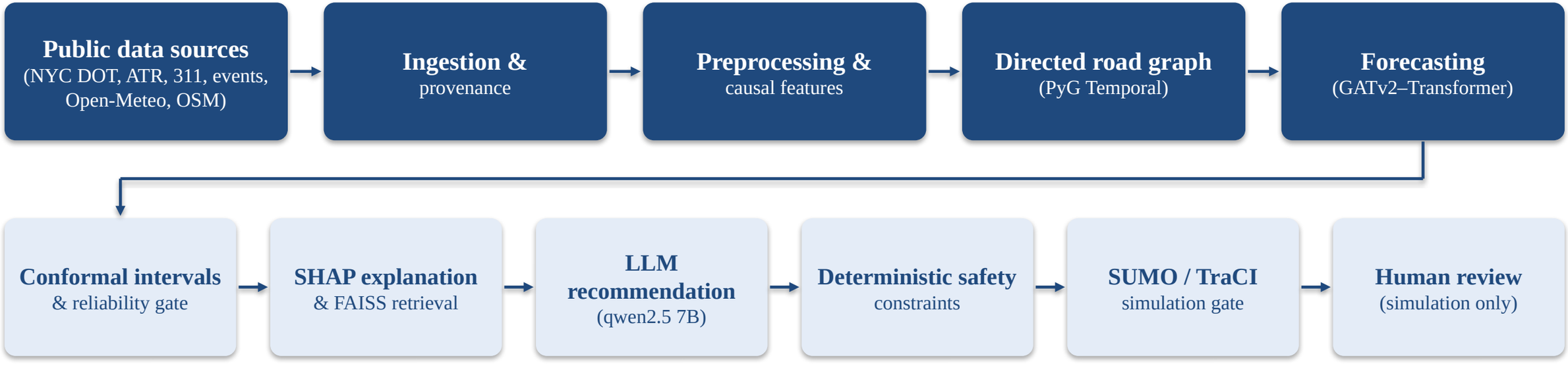
### M9 Safety constraints
13 deterministic checks; protected TraCI signal controller

### M10 Simulation
SUMO 1.28 West Street corridor; matched seeds; testbed audit

### M11 Services
FastAPI, PostgreSQL/PostGIS job queue, worker, Streamlit dashboard

### M12 Tracking & reporting
MLflow, evidence registry, claims ledger, paper tables from artifacts

# SYSTEM ARCHITECTURE




## Data flow

Raw API responses stored verbatim with hashes → 5-minute grid per link → windows of 24 steps × nodes × features → forecasts with 90 % intervals → advisory state in LangGraph → decision and audit trail persisted in PostgreSQL/PostGIS; every run logged in MLflow.

## Technology stack

Python 3.12 · Pandas/Polars · PyTorch + PyTorch Geometric + PyG Temporal · XGBoost/LightGBM · SHAP · sentence-transformers + FAISS · Ollama (qwen2.5:7b) · LangGraph · SUMO 1.28 / TraCI · FastAPI · Streamlit · PostgreSQL/PostGIS · MLflow · Docker Compose · Pytest

# DEVELOPMENT & INTEGRATION

## Implementation of modules

One Python package with a staged command-line interface, YAML configuration and Pydantic contracts

Every stage writes artifacts with run IDs and SHA-256 hashes

The same code path runs the smoke tests and the full experiment

## Integration strategy

Shared contracts frozen first: forecast, uncertainty, evidence, candidate action, decision

A stage runner executes the real-data pipeline in fixed order with a stage log

Only the recommendation node calls the LLM; every gate is deterministic code

95 automated tests: unit, leakage, safety, integration, end-to-end (CI, 2026-10-25)

## Challenges faced → solutions

**7 of 26 links report zero speed all year**
→ treated as missing; 16 links kept by a train-coverage rule

**DOT link geometries stop short of link ends**
→ three documented graph-edge rules

**GRU crash with cuDNN on Windows/CUDA**
→ cuDNN disabled for GRU runs only

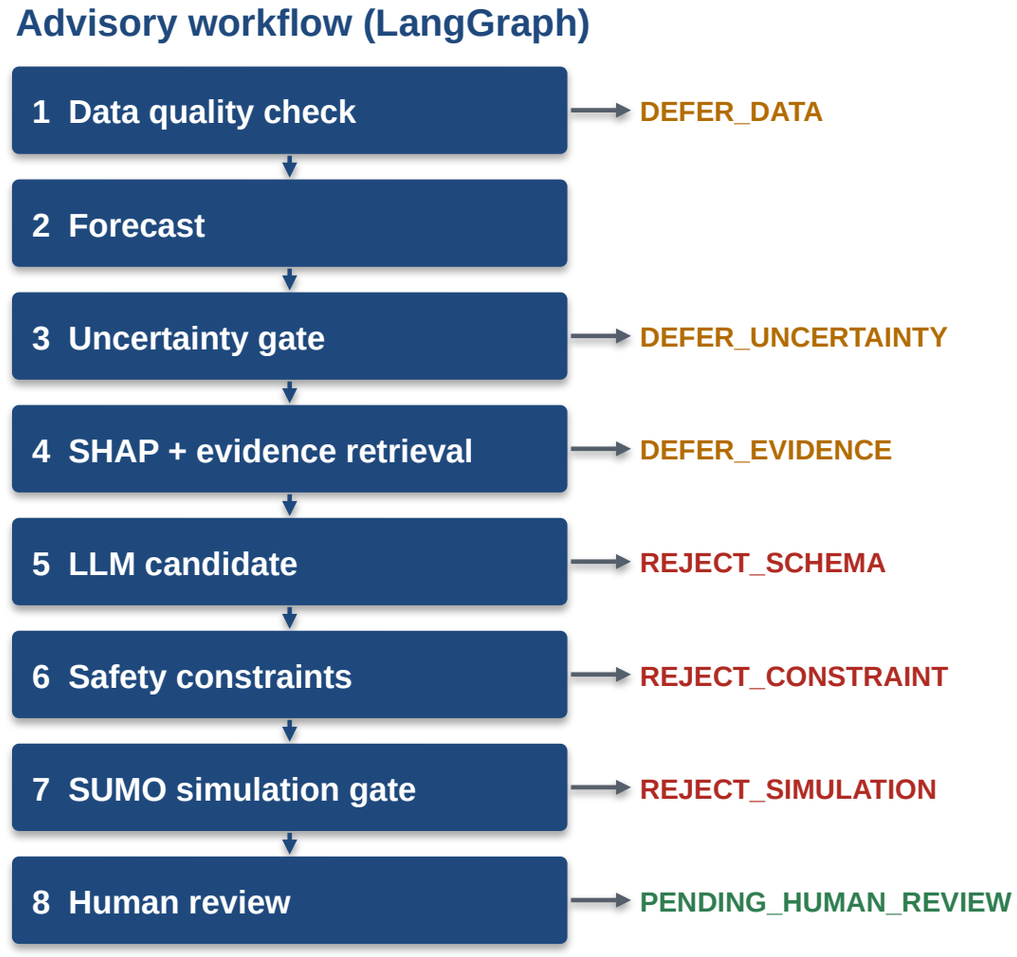
**Bootstrap resampled 5-minute steps, not days**
→ unit fixed and ablations re-run

**A missing field silently blocked every LLM call**
→ fixed and agent evaluation re-run

**SUMO re-routed half the corridor traffic**
→ fixed routes and a testbed audit

# PROTOTYPE / RESEARCH FRAMEWORK

- **Command-line pipeline:** one command per stage, from data audit to paper tables
- **FastAPI service:** 18 endpoints for forecasts, explanations, retrieval, advisories, simulations and review; long jobs run in a PostgreSQL-backed worker
- **Streamlit dashboard:** 11 pages, from data audit and traffic map to forecasts, explanations, agent traces, SUMO results and the model card
- **Every view labels its mode:** LIVE, REPLAY or SIMULATION
- **Docker Compose:** PostGIS, API, worker and dashboard images, built and smoke-tested
- **Human review** approves or rejects for simulation only; nothing is sent to real signals



# 7. CONCLUSIONS

## Performance metrics (test period 6 Sep – 20 Oct 2026)

| Metric | 15 min | 30 min | 60 min |
|---|---|---|---|
| Proposed model MAE (mph) | **4.77** | **5.33** | **6.15** |
| Proposed, mean of 3 seeds (mph) | 4.69 | 5.34 | 6.15 |
| Persistence MAE (mph) | 5.14 | 6.06 | 7.29 |
| 90 % interval coverage | 0.909 | 0.907 | 0.903 |
| Coverage under congestion | 0.798 | 0.794 | 0.757 |
| MAE kept after gate (≈26 % deferred) | 3.72 | 4.38 | 5.25 |

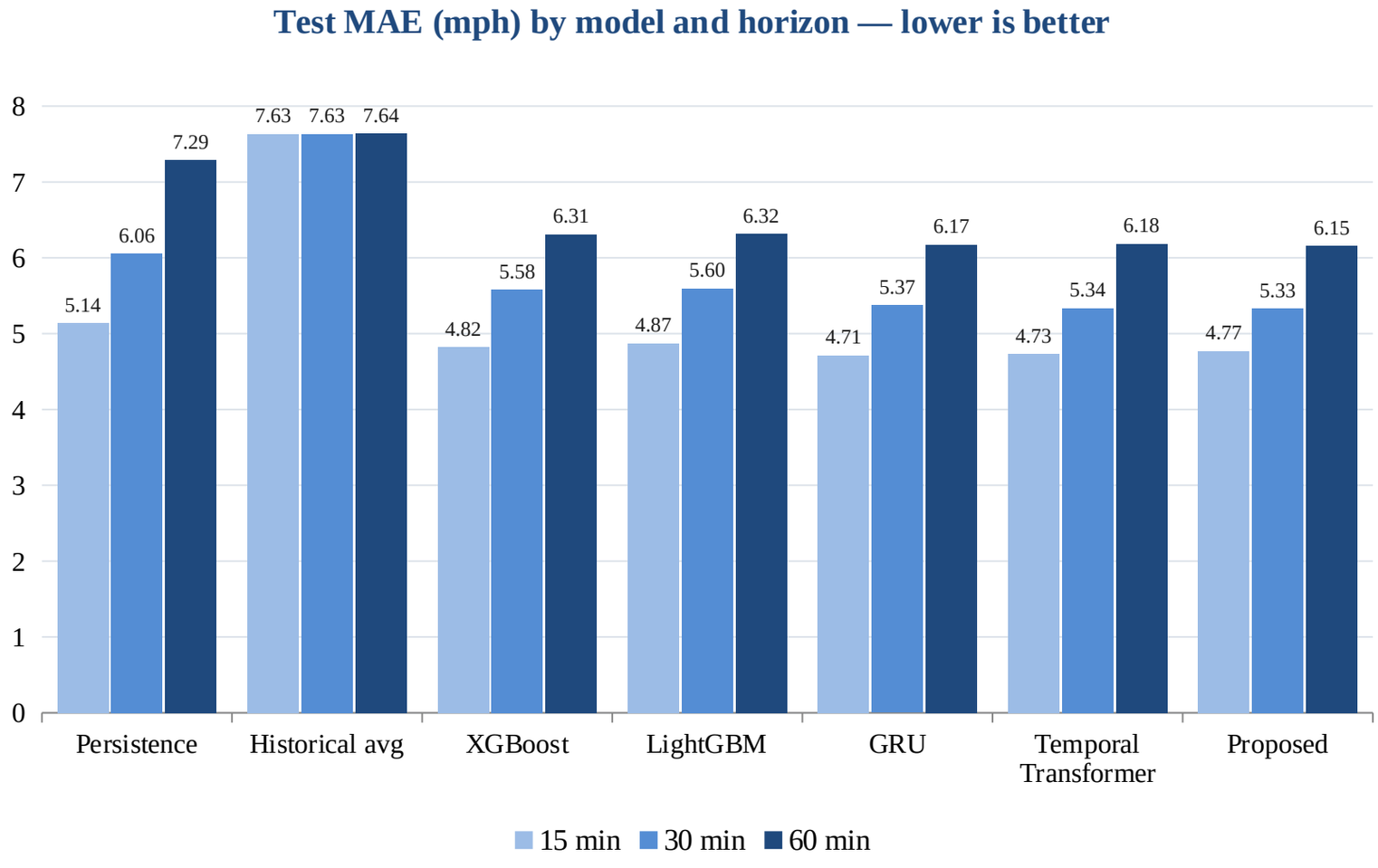
## Validation of approach

Chronological splits with purge gaps and leakage tests; identical targets and masks for every model; day-block bootstrap for ablations; a preregistered simulation baseline test; failed runs kept and reported.

## Observations

- The proposed model beats persistence by 7–16 % and both tree models. Over three seeds it is 0.03–0.08 mph below the temporal Transformer at every horizon and the GRU at 30/60 min; the GRU is better at 15 min. Small, not decisive.
- The availability-aware gate helps (MAE +0.067 mph when removed); operational weather features hurt (−0.046 mph), a negative result.
- Intervals meet 90 % overall but under-cover congestion, the regime that matters for advisories.
- Retrieval removed fabricated citations (3 of 3 without it, 0 of 9 with it); the constraint layer, not the LLM, decides admissibility.
- The simulation testbed still fails its own baseline test after signal-split calibration (8.1 vs 10.95 mph), so the controller comparison is not run.

# 8. OUTCOME



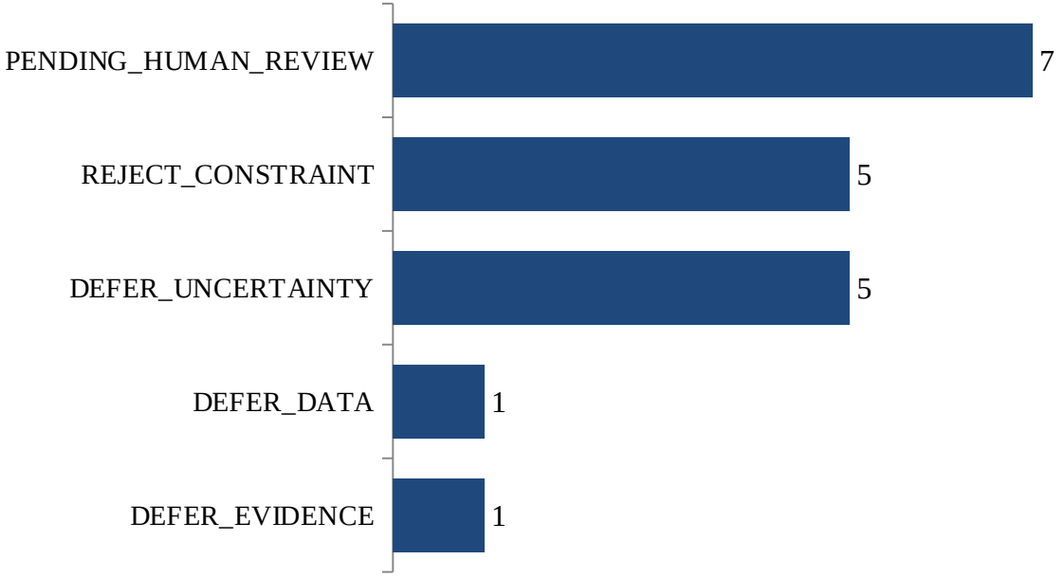

## Key results

- Every learned model beats persistence and the historical average.
- Proposed model: 4.77 / 5.33 / 6.15 mph; lowest at 30 and 60 min, GRU lowest at 15 min.
- $R^2$ rises from 0.55 / 0.46 / 0.30 (persistence) to 0.69 / 0.63 / 0.56.
- 144,205–144,217 test targets per horizon, identical for every model.

Data: NYC DOT speeds [11], 1 Jan – 20 Oct 2026.

# 8. OUTCOME (CONTD.)

### Advisory decisions in 19 workflow runs (REPLAY cases)




12 of 12 LLM responses were schema-valid. Without retrieval the model cited non-existent evidence in 3 of 3 cases (all rejected); with retrieval in 0 of 9.

### SUMO testbed outcome (simulation only)

| Protocol | Unfinished* | Result |
|---|---|---|
| First comparison, 20 Oct (45 runs) | — | **Withdrawn** |
| 2014 count, generated signal timing | 17.7 % | **Fails test** |
| 2014 count, Webster re-timed signals | 4.3 % | **Fails test** |
| 2025 counts, Webster, no turn-lane limits | 10.1 % | **Fails test** |
| v5: 2025 counts, corridor green ×1.0–2.5 | 0–0.5 % at ×2.0+ | **8.1 of 10.95 mph** |

* Vehicles not finishing with no side-street traffic; the preregistered test allows at most 1 % and no stuck vehicles.

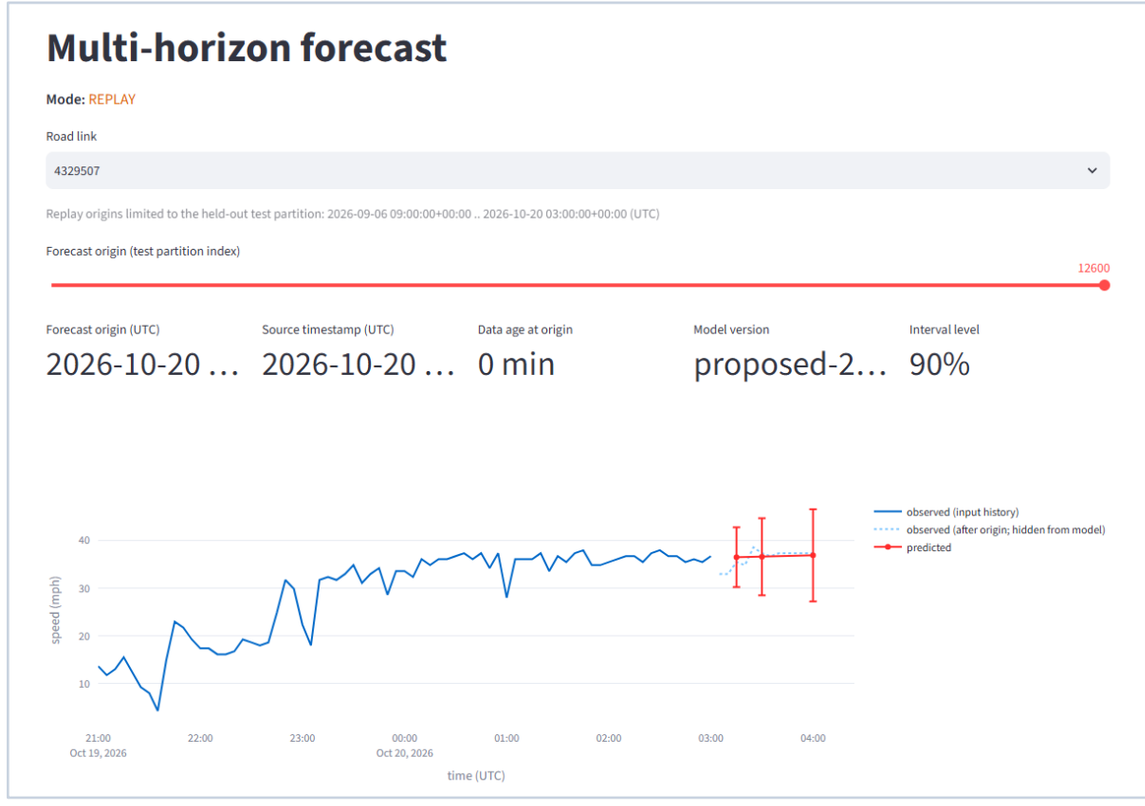
The first comparison let SUMO re-route 1,120 of 2,189 main-road vehicles off West Street, so it did not test the corridor.

With real 2025 counts (2,510 NB / 2,444 SB veh/h) the modelled link runs at 5.8 mph; the real street runs at 10.95 mph in the PM peak.

**v5 (23 Oct, 21 runs): more corridor green raises it to 8.1 mph, still 2.85 mph short of the 1.5 mph tolerance. No timing selected; the comparison stays NOT RUN.**

# 8. OUTCOME: WORKING DASHBOARD (1/5)

## Multi-horizon forecast (proposed model) Mode shown: **REPLAY**




## Traffic map: monitored links Mode shown: **REPLAY**




Captured on 23 Oct 2026 on the training laptop, which holds the trained model and the processed NYC DOT data. The forecast is a REPLAY of the held-out test period (origin 20 Oct 2026, 90 % conformal intervals); no LIVE view is shown.

# 8. OUTCOME: WORKING DASHBOARD (2/5)

## Overview & status (status page, no mode label)

### Multi-horizon traffic forecasting & explainable agentic signal advisory

Predictive Analytics for Smart City Infrastructure - Multi-Horizon Traffic Congestion Forecasting and Explainable Agentic Signal Advisory Using Spatiotemporal Transformers and RAG

| Proposed model | Track | Study area |
|---|---|---|
| proposed-20261020T… | operational | manhattan_dot_spee… |

#### Component status (evidence registry)

| requirement_id | component | test_status | artifact_path | limitations |
|---|---|---|---|---|
| DATA-01 | data | VERIFIED | reports/data_audit/source_inventory.csv | None |
| DATA-02 | data | PARTIAL | data/manifests/source_manifest.json | The forecasting data (sp |
| DATA-03 | data | VERIFIED | reports/data_audit/availability_audit.md | None |
| GRAPH-01 | graph | VERIFIED | reports/data_audit/graph_validation.json | None |
| GRAPH-02 | graph | VERIFIED | artifacts/models/proposed-20261020T210211-4a0685/feature_schema.json | None |
| MODEL-01 | models | VERIFIED | artifacts/models/registry_operational.json | None |
| MODEL-02 | models | VERIFIED | reports/experiments/forecast_metrics.csv | None |
| UNC-01 | uncertainty | VERIFIED | artifacts/calibration/calibrator.json | None |
| XAI-01 | explainability | VERIFIED | artifacts/explanations/explanations_summary.json | None |
| RAG-01 | rag | VERIFIED | artifacts/retrieval/retrieval_eval.json | None |

› final_status.md

## Data audit & provenance Mode shown: REPLAY

### Data audit & provenance

**Mode:** REPLAY

| source_id | official_title | publisher | dataset_id | endpoint |
|---|---|---|---|---|
| speeds | DOT Traffic Speeds | City of New York | i4gi-tjb9 | https://data.cityofnewyork.us/resource/i4gi-tjb9.json |
| volume | Automated Traffic Volume Counts | City of New York | 7ym2-wayt | https://data.cityofnewyork.us/resource/7ym2-wayt.json |
| events | NYC Permitted Event Information | Office of Citywide Event Coord | tvpp-9vvx | https://data.cityofnewyork.us/resource/tvpp-9vvx.json |
| events_hist | NYC Permitted Event Information - Historical | City of New York | bkfu-528j | https://data.cityofnewyork.us/resource/bkfu-528j.json |
| collisions | Motor Vehicle Collisions - Crashes | Police Department (NYPD) | h9gi-nx95 | https://data.cityofnewyork.us/resource/h9gi-nx95.json |
| sr311 | 311 Service Requests from 2020 to Present | 311 | erm2-nwe9 | https://data.cityofnewyork.us/resource/erm2-nwe9.json |
| weather | Open-Meteo archive (ERA5-based reanalysis) + historical forecast API | Open-Meteo | | https://archive-api.open-meteo.com/v1/archive;https://historic |
| osm | OpenStreetMap extracts via Overpass API | OpenStreetMap contributors | | https://overpass-api.de/api/interpreter |

#### Coverage by year (server-side counts)

| source_id | dataset_id | year | record_count |
|---|---|---|---|
| speeds | i4gi-tjb9 | 1930 | 8 |
| speeds | i4gi-tjb9 | 1978 | 15 |
| speeds | i4gi-tjb9 | 2014 | 6 |
| speeds | i4gi-tjb9 | 2015 | 9 |
| speeds | i4gi-tjb9 | 2016 | 36 |
| speeds | i4gi-tjb9 | 2017 | 6019989 |
| speeds | i4gi-tjb9 | 2018 | 13860107 |
| speeds | i4gi-tjb9 | 2019 | 12971855 |
| speeds | i4gi-tjb9 | 2020 | 11943235 |
| speeds | i4gi-tjb9 | 2021 | 12203801 |

Screenshots of the project's Streamlit dashboard (src/smartcity/dashboard/app.py) captured on 21 Oct 2026; every number on them is read from committed result files. Live NYC feeds were not reachable from the capture machine, so no LIVE view is shown.

# 8. OUTCOME: WORKING DASHBOARD (3/5)

## Experiment results Mode shown: REPLAY

### Experiment results

**Mode:** REPLAY

#### forecast_metrics

| run_id | model | track | split | evaluation_period_utc | horizon_min | n_valid_targets | mae_mp |
|---|---|---|---|---|---|---|---|
| persistence-20261020T193923-c48849 | persistence | operational | test | 2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00 | 15 | 144217 | 5.13860 |
| persistence-20261020T193923-c48849 | persistence | operational | test | 2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00 | 30 | 144213 | 6.05562 |
| persistence-20261020T193923-c48849 | persistence | operational | test | 2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00 | 60 | 144205 | 7.28756 |
| historical_average-20261020T193932-e08975 | historical_average | operational | test | 2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00 | 15 | 144217 | 7.63218 |
| historical_average-20261020T193932-e08975 | historical_average | operational | test | 2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00 | 30 | 144213 | 7.63295 |
| historical_average-20261020T193932-e08975 | historical_average | operational | test | 2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00 | 60 | 144205 | 7.63695 |
| xgboost-20261020T193933-18d124 | xgboost | operational | test | 2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00 | 15 | 144217 | 4.82294 |
| xgboost-20261020T193933-18d124 | xgboost | operational | test | 2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00 | 30 | 144213 | 5.57922 |
| xgboost-20261020T193933-18d124 | xgboost | operational | test | 2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00 | 60 | 144205 | 6.30799 |
| lightgbm-20261020T193956-ccbf73 | lightgbm | operational | test | 2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00 | 15 | 144217 | 4.87253 |

#### ablation_metrics

| variant | track | run_id | mae_mean_mph | delta_vs_proposed_mph | ci95_low | ci95_high | n_days | status |
|---|---|---|---|---|---|---|---|---|
| proposed | operational | proposed-20261020T201110-0e333f | 5.4194 | 0.0 | 0.0 | 0.0 | 45 | REFERENC |
| ablation_no_gate | operational | ablation_no_gate-20261020T202053-48c9ff | 5.4867 | 0.0673 | 0.0482 | 0.0864 | 44 | OK |
| ablation_no_weather | operational | ablation_no_weather-20261020T202610-295c01 | 5.3732 | -0.0462 | -0.0692 | -0.0205 | 44 | OK |
| ablation_no_infrastructure | operational | ablation_no_infrastructure-20261020T203545-6cd844 | 5.433 | 0.0136 | 0.0019 | 0.0258 | 44 | OK |
| ablation_no_events | operational | ablation_no_events-20261020T204114-372604 | | | | | | NOT_APPL |

## Uncertainty & reliability Mode shown: REPLAY

#### coverage_by_horizon.csv

| horizon_min | group | n | coverage | mean_width_mph |
|---|---|---|---|---|
| 15 | all | 144217 | 0.909 | 24.2562 |
| 30 | all | 144213 | 0.9074 | 26.945 |
| 60 | all | 144205 | 0.9033 | 29.5394 |

#### coverage_by_scenario.csv

| horizon_min | group | n | coverage | mean_width_mph |
|---|---|---|---|---|
| 15 | free_ratio>=0.8 | 98556 | 0.9311 | 24.2562 |
| 15 | moderate_0.5-0.8 | 22396 | 0.9266 | 24.2562 |
| 15 | congested<0.5 | 23265 | 0.7982 | 24.2562 |
| 30 | free_ratio>=0.8 | 98546 | 0.9301 | 26.945 |
| 30 | moderate_0.5-0.8 | 22398 | 0.9259 | 26.945 |
| 30 | congested<0.5 | 23269 | 0.7935 | 26.945 |
| 60 | free_ratio>=0.8 | 98522 | 0.9329 | 29.5394 |
| 60 | moderate_0.5-0.8 | 22406 | 0.9253 | 29.5394 |
| 60 | congested<0.5 | 23277 | 0.7569 | 29.5394 |
| 15 | non_rush | 106048 | 0.8984 | 24.2562 |

#### deferral_tradeoff.csv

| missing_threshold | horizon_min | n_valid | retained | deferral_rate | mae_retained | mae_all | coverage_retained |
|---|---|---|---|---|---|---|---|
| 0 | 15 | 144217 | 102928 | 0.2863 | 3.5605 | 4.7727 | 0.9531 |

Screenshots of the project's Streamlit dashboard (src/smartcity/dashboard/app.py) captured on 21 Oct 2026; every number on them is read from committed result files. Live NYC feeds were not reachable from the capture machine, so no LIVE view is shown. The Experiment results page shows the seed-42 runs; the three-seed results are on slide 22.

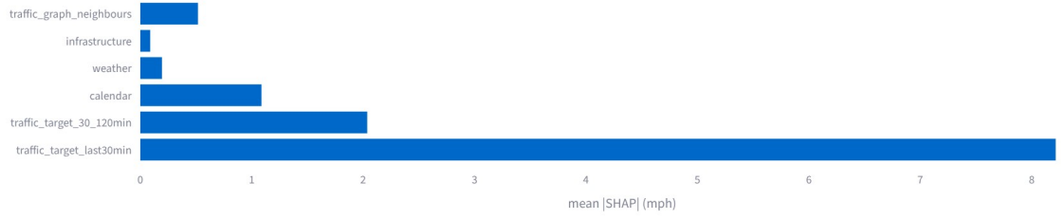
# 8. OUTCOME: WORKING DASHBOARD (4/5)

## Explainability (proposed model) Mode shown: REPLAY

### Explanations of the proposed model

**Mode:** REPLAY

proposed (context-gated GATv2-Transformer) — grouped KernelSHAP; sensitivities, not causal effects.




Local alert

4620298 h=30


```
{
  "explanation_id" : "xai-20261020T212531-006725"
  "method" : "grouped KernelSHAP (exact, all coalitions) on the proposed model"
  "link_id" : "4620298"
  "link_index" : 15
  "horizon_minutes" : 30
  "origin_index" : 83417
  "origin_utc" : 1792269000
  "background" : "16 windows sampled from TRAIN partition (seed 42)"
  "players" : [
```


## Agent trace & advisory Mode shown: REPLAY

### Agent traces & advisories

| run_id | case | decision | reasons |
|---|---|---|---|
| adv-20261021T043410-63bde3 | mapped_corridor_rush_hour | PENDING_HUMAN_REVIEW | monitor_only advisory (no signal change proposed) |
| adv-20261021T043340-e5677a | rain_period | DEFER_UNCERTAINTY | WIDE_INTERVAL: h=15 width 44.4 > 38.1 mph; WIDE_INTERVAL: h=30 width 45.8 > 38.1 m |
| adv-20261021T043340-9374e9 | mapped_corridor_alert | PENDING_HUMAN_REVIEW | monitor_only advisory (no signal change proposed) |
| adv-20261021T043316-927b0c | routine_rush_hour | REJECT_CONSTRAINT | network_mapping_valid: no reviewed DOT->SUMO mapping |
| adv-20261021T043252-38b42e | mapped_corridor_rush_hour | PENDING_HUMAN_REVIEW | monitor_only advisory (no signal change proposed) |
| adv-20261021T043216-ea1e44 | rain_period | DEFER_UNCERTAINTY | WIDE_INTERVAL: h=15 width 44.4 > 38.1 mph; WIDE_INTERVAL: h=30 width 45.8 > 38.1 m |
| adv-20261021T043216-99095f | mapped_corridor_alert | PENDING_HUMAN_REVIEW | monitor_only advisory (no signal change proposed) |
| adv-20261021T043143-66b3d2 | routine_rush_hour | PENDING_HUMAN_REVIEW | monitor_only advisory (no signal change proposed) |
| adv-20261021T043115-bf3109 | mapped_corridor_rush_hour | REJECT_CONSTRAINT | retrieval_support_acceptable: evidence ids not in retrieved set: ['EVIDENCE_123456'] |
| adv-20261021T043049-b4865d | mapped_corridor_alert | REJECT_CONSTRAINT | retrieval_support_acceptable: evidence ids not in retrieved set: ['EVIDENCE_123456'] |

Trace

adv-20261021T043410-63bde3

**Mode:** REPLAY

**Decision:** `PENDING_HUMAN_REVIEW`

- data_quality
- forecast
- uncertainty
- retrieval

Screenshots of the project's Streamlit dashboard (src/smartcity/dashboard/app.py) captured on 21 Oct 2026; every number on them is read from committed result files. Live NYC feeds were not reachable from the capture machine, so no LIVE view is shown. Agent traces end in review, deferral or rejection; nothing is sent to real signals.

# 8. OUTCOME: WORKING DASHBOARD (5/5)

## SUMO comparisons Mode shown: SIMULATION

### SUMO comparisons

**Mode:** SIMULATION

Scenario-based microscopic simulation testbed using real-world road geometry (OSM) and declared demand assumptions. Not a calibrated digital twin; not real-world gains.

**SUPERSEDED (C12)** Matched-seed SUMO comparison of baseline vs protected controller (2026-10-20, 45 runs): 45 runs completed as recorded, but the 2026-10-21 testbed audit (simulation_testbed_audit.json) found they used bare from/to flows that SUMO re-routes at insertion (1,120 of 2,189 mainline routes left West Street) and a 0.2 m northbound entry stub, so they are not a corridor controller comparison; kept as evidence, no controller ranking claimed

**Controller comparison: NOT RUN.** (C31) With 2025 counts, no turn-lane restrictions and Webster programs no level passes the rule (523-568 of 5,368 unfinished with no side demand); NB completes 2,520-2,524 trips, SB 1,877-1,922; one SB signal estimate 2,278 < 2,444 veh/h; 15 runs, fixed-time only; comparison stays NOT RUN.

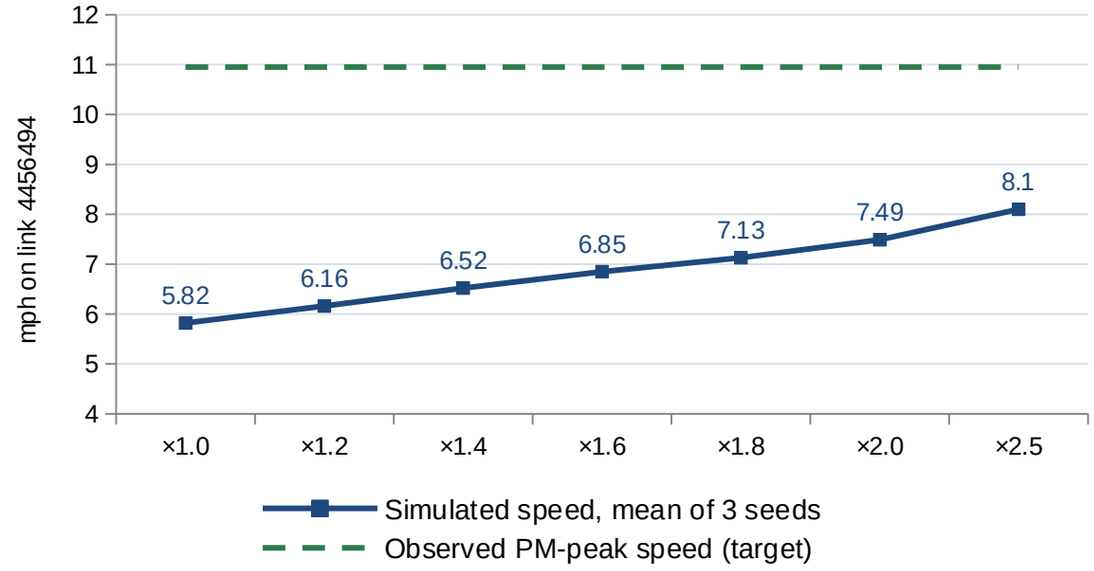
**Testbed audit: preregistered demand-selection runs (fixed-time only)**

v4_counts2025_webster_noturnlanes (15 runs)

| scenario | seed | main_nb_vph | main_sb_vph | side_vph_per_approach | loaded | completed_trips | unfinished_trips | teleports | corridor_mean_time_loss_s |
|---|---|---|---|---|---|---|---|---|---|
| sweep_side100 | 101 | 2510 | 2444 | 150 | 9443 | 6509 | 2254 | 70 | 914.7928 |
| sweep_side100 | 102 | 2510 | 2444 | 150 | 9443 | 6592 | 2172 | 49 | 918.8012 |
| sweep_side100 | 103 | 2510 | 2444 | 150 | 9443 | 6557 | 2210 | 78 | 925.3402 |
| sweep_side075 | 101 | 2510 | 2444 | 112.5 | 8418 | 6428 | 1380 | 50 | 746.9846 |
| sweep_side075 | 102 | 2510 | 2444 | 112.5 | 8418 | 6374 | 1436 | 54 | 748.6056 |
| sweep_side075 | 103 | 2510 | 2444 | 112.5 | 8418 | 6339 | 1471 | 50 | 760.6718 |
| sweep_side050 | 101 | 2510 | 2444 | 75 | 7418 | 5868 | 1000 | 42 | 701.0772 |
| sweep_side050 | 102 | 2510 | 2444 | 75 | 7418 | 5872 | 997 | 41 | 684.0547 |
| sweep_side050 | 103 | 2510 | 2444 | 75 | 7418 | 5873 | 995 | 49 | 694.5947 |
| sweep_side025 | 101 | 2510 | 2444 | 37.5 | 6393 | 5179 | 726 | 29 | 675.6254 |

Captured 21 Oct 2026 from committed result files; no LIVE view.

## Signal-split calibration, protocol v5 Mode: SIMULATION




- Corridor green share scaled ×1.0–2.5 (21 fixed-time runs, seeds 101–103, 2025 counts, 23 Oct 2026).
- Admissible only at ×2.0 and ×2.5 (no teleports, at most 28 of 5,368 trips unfinished); ×1.0–1.8 leave 98–568 unfinished.
- **Best admissible 8.10 mph is 2.85 mph below 10.95, beyond the 1.5 mph tolerance: no timing selected, controller comparison NOT RUN, rule not relaxed.**

Run 23 Oct, after the capture; from simulation_split_calibration_v5_*.csv

# 9. RESULTS & OUTCOMES

Held-out test period 6 Sep – 20 Oct 2026, 16 Manhattan links; advisory and retrieval results from the REPLAY workflow runs

## Test MAE (mph): mean ± s.d. over seeds 42, 1 and 2

| Model | 15 min | 30 min | 60 min |
|---|---|---|---|
| **Proposed GATv2–Transformer** | **4.69 ± 0.06** | **5.34 ± 0.03** | **6.15 ± 0.02** |
| GRU | 4.67 ± 0.06 | 5.37 ± 0.02 | 6.18 ± 0.01 |
| Temporal Transformer | 4.72 ± 0.03 | 5.38 ± 0.03 | 6.23 ± 0.08 |
| Proposed, seed 42 only | 4.77 | 5.33 | 6.15 |
| Persistence | 5.14 | 6.06 | 7.29 |

Paired day-block bootstrap (44 test days): the proposed model is lower than the temporal Transformer at every horizon (0.035 / 0.037 / 0.080 mph) and than the GRU at 30 and 60 min (0.032 / 0.031); the GRU is lower at 15 min (0.021). All intervals exclude zero, but with three seeds the advantage is small, not decisive.

Source: reports/experiments/multi_seed_summary.csv (nine GPU runs, 21 Oct 2026).

**7–16 %**
lower MAE than persistence (seed 42 run)

**0.91 / 0.91 / 0.90**
coverage of the 90 % conformal intervals; 0.76–0.80 under congestion

**15–22 %**
lower MAE on forecasts the reliability gate keeps (≈26 % deferred)

**24 / 24**
local-LLM responses schema-valid over two workflow passes (qwen2.5 7B)

**3 of 3 → 0 of 9**
advisories citing non-existent evidence, without vs with retrieval

**19**
workflow runs per pass: 7 pending human review, 5 rejected by safety checks, 7 deferred

**Also measured:** removing the availability-aware gate raises MAE by 0.067 mph (95 % CI 0.048–0.086); removing operational weather features lowers it by 0.046 mph (a negative result). No simulated controller gain is claimed: the SUMO testbed fails its own baseline test (8.1 vs 10.95 mph after signal-split calibration).

# 9. RESULTS & OUTCOMES (CONTD.)

## Implemented and verified

- Real-data pipeline on NYC DOT speeds, 1 Jan – 20 Oct 2026, with provenance hashes
- Seven forecasting models on identical targets; three-seed study of the deep models; ablations
- Conformal intervals, reliability gate and grouped SHAP on the trained model
- FAISS retrieval over 15 approved documents; local LLM with schema validation
- LangGraph workflow with 13 deterministic safety checks
- FastAPI, PostGIS jobs, Streamlit, MLflow, Docker Compose

## Partial or not run

- SUMO controller comparison NOT RUN: signal-split calibration (v5, 23 Oct) reaches 8.1 of 10.95 mph; the first 45-run comparison is superseded
- No real advisory has reached the simulation gate (the LLM chose monitor-only in both passes)
- Only three seeds; the proposed model is not bit-reproducible on GPU at a fixed seed
- Road graph is sparse: 26 nodes, 16 edges
- Intervals under-cover congested periods

## Outcomes and next steps

- IEEE conference paper (LaTeX and Word), every claim traced in the claims ledger
- GitHub repository with tests, runbooks and result files
- Next: run preregistered protocol v6 (cycle × green × offsets, 75 runs; prepared, not run), then the comparison
- Next: more seeds without the compute budget
- Next: paper submission

# REFERENCES


[1] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in Proc. ICLR, 2018.

[2] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in Proc. IJCAI, 2018, pp. 3634–3640.

[3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, “Graph WaveNet for deep spatial-temporal graph modeling,” in Proc. IJCAI, 2019, pp. 1907–1913.

[4] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Proc. NeurIPS, 2020.

[5] P. A. Lopez et al., “Microscopic traffic simulation using SUMO,” in Proc. IEEE ITSC, 2018, pp. 2575–2582.

[6] S. Brody, U. Alon, and E. Yahav, “How attentive are graph attention networks?,” in Proc. ICLR, 2022.

[7] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” arXiv:2107.07511, 2021.

[8] C. Xu and Y. Xie, “Conformal prediction interval for dynamic time-series,” in Proc. ICML, PMLR vol. 139, 2021, pp. 11559–11569.

[9] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Proc. NeurIPS, 2017.

[10] B. Rozemberczki et al., “PyTorch Geometric Temporal: Spatiotemporal signal processing with neural machine learning models,” in Proc. ACM CIKM, 2021, pp. 4564–4573.

[11] NYC Department of Transportation, “DOT Traffic Speeds NBE,” NYC Open Data, dataset i4gi-tjb9, accessed Oct. 2026.

[12] NYC Department of Transportation, “Automated Traffic Volume Counts,” NYC Open Data, dataset 7ym2-wayt, accessed Oct. 2026.

# Thank You